The Black Hole Information Paradox Resolved: Ledger Conservation in Recognition Science

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Abstract

We resolve the black hole information paradox within Recognition Science (RS). The resolution rests on three structural properties of the recognition ledger: (i) reciprocal symmetry ($J(x) = J(x^{-1})$) forces double-entry bookkeeping, so every ledger entry has a permanent counterpart and no entry can be deleted; (ii) the ledger is append-only, yielding a no-sink theorem: no operation reduces the total recognition charge Z_{total}; (iii) the event horizon is a causal boundary on the ledger, not an information barrier—entries inside the horizon remain part of the global ledger and influence future states through J-cost correlations. The Bekenstein–Hawking entropy $S_{\text{BH}} = A/(4\ell_P^2)$ counts the number of independent φ-bits on the horizon surface. The Hawking temperature $T_H = \hbar c^3/(8\pi G M k_B)$ is the J-cost of the lowest-rung excitation at the horizon. The classical information loss occurs only in the semiclassical approximation, where the ledger's discrete correlations (double-entry pairs) are replaced by a smooth manifold that cannot encode them. When the full discrete structure is retained, information is manifestly conserved and the Page curve is reproduced.

1 Introduction

Hawking's 1976 calculation [3] showed that black holes radiate thermally at temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}, \tag{1}$$

where M is the black hole mass. Since a thermal spectrum is maximally entropic, it carries no information about the matter that formed the black hole. If the black hole evaporates completely, the original quantum state is lost—violating unitarity.

This "information paradox" has generated decades of debate. Proposed resolutions include:

1. **Subtle correlations**: Information escapes in Hawking radiation via late-time correlations (Page curve [4]).
2. **Remnants**: The evaporation endpoint retains the information.

3. **Complementarity**: Information is simultaneously inside and outside the horizon [5].
4. **Firewall**: Unitarity requires a high-energy surface at the horizon [6].
5. **ER=EPR**: Entanglement is geometric connectivity [7].
6. **Quantum extremal surfaces**: Island formula recovers the Page curve holographically [8].

RS resolves the paradox from first principles [2]: the ledger conserves information by construction, without invoking holography or new spacetime topology.

2 The Cost Functional and Ledger Structure

Definition 2.1 (RS Cost Functional)**.** For $x > 0$:

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1. \tag{2}$$

The functional J is uniquely determined by three axioms:
Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

Lemma 2.3 (Reciprocal Symmetry)**.** $J(x) = J(x^{-1})$ *for all* $x > 0$.

Proof. $J(x^{-1}) = \frac{1}{2}(x^{-1} + x) - 1 = \frac{1}{2}(x + x^{-1}) - 1 = J(x)$. □

Lemma 2.4 (Unique Minimum)**.** $J(x) \geq 0$ *with equality if and only if* $x = 1$.

Proof. $J(x) = \frac{1}{2}(x + x^{-1}) - 1 \geq \frac{1}{2} \cdot 2\sqrt{x \cdot x^{-1}} - 1 = 0$ by the AM-GM inequality, with equality iff $x = x^{-1}$, i.e., $x = 1$. □

3 Double-Entry Conservation

Proposition 3.1 (Double-Entry Bookkeeping — Structural Postulate)**.** *The ledger operates on strict double-entry bookkeeping: every recognition event* e_i *with ratio* x_i *is paired with its inverse* e_i^{-1} *with ratio* x_i^{-1}*. No entry can be created without its counterpart, and no entry can be deleted.*

Structural argument. The Recognition Composition Law (RCL) treats x and x^{-1} symmetrically via Lemma 2.3: the cost of a process equals the cost of its inverse. This structural symmetry motivates the double-entry interpretation. The append-only property (no entry is ever removed from the ledger) is a foundational postulate of the RS framework [2]: the ledger grows monotonically in tick number, and no retroactive deletion of ledger entries is permitted. The double-entry pairing is a consequence of the $J(x) = J(x^{-1})$ symmetry applied as a bookkeeping principle: every debit has a credit. A derivation of the append-only property from a more primitive axiom—rather than as a postulate—is an identified open problem. □

4 The No-Sink Theorem

Definition 4.1 (Recognition Charge)**.** Each recognition event e_i carries recognition charge $Z_i \geq 0$. The total recognition charge of the ledger is $Z_{\text{total}} = \sum_i Z_i$.

Theorem 4.2 (No Information Sink)**.** *The recognition ledger has no information sinks: no ledger operation reduces Z_{total}.*

Proof. Recognition events can be created (in double-entry pairs) or redistributed among ledger sectors, but never annihilated. Annihilation of event e_i would require removing both e_i and e_i^{-1}, reducing Z_{total} by $2Z_i > 0$. But removal is not a valid ledger operation: the ledger is append-only. Therefore $Z_{\text{total}}(t+1) \geq Z_{\text{total}}(t)$ for all ticks t. □

Corollary 4.3 (Information Conservation)**.** *The ledger information content $I = \sum_i \ln(1+Z_i/\varphi)/\ln\varphi$ (total φ-bits) satisfies $I(t+1) \geq I(t)$. Information is monotonically non-decreasing.*

5 The Event Horizon as a Causal Boundary

Theorem 5.1 (Horizon Is Not an Information Barrier)**.** *The event horizon is a causal boundary (restricting signal propagation) but not an information barrier (ledger entries inside the horizon remain part of the global ledger).*

Proof. The ledger is a global structure: all recognition events, regardless of spatial location, are entries in the same ledger. The horizon restricts *causal influence*—which events can signal which others, determined by the light-cone structure (one voxel per tick, c). It does not restrict *informational existence*: an event inside the horizon remains a ledger entry, contributes to Z_{total}, and retains its double-entry partner.

Formally: let $\mathcal{L}_{\text{in}}$ and $\mathcal{L}_{\text{out}}$ be the sets of ledger entries inside and outside the horizon. Then $Z_{\text{total}} = Z_{\text{in}} + Z_{\text{out}}$, with $Z_{\text{in}} \geq 0$ and $Z_{\text{out}} \geq 0$. The causal boundary prevents $\mathcal{L}_{\text{in}}$ from *signaling* $\mathcal{L}_{\text{out}}$, but does not remove $\mathcal{L}_{\text{in}}$ from the ledger. □

Remark 5.2. In semiclassical GR, informational existence and causal accessibility are conflated because information is identified with signal propagation. In RS, information is a ledger property, independent of signaling.

6 Bekenstein–Hawking Entropy

Theorem 6.1 (Area-Entropy Relation)**.** *The entropy of a black hole with horizon area A is:*

$$\boxed{S_{\text{BH}} = \frac{A}{4\,\ell_P^2},} \tag{3}$$

where $\ell_P = \sqrt{\hbar G/c^3}$ is the Planck length.

Proof sketch. The horizon surface is tiled by Planck-area cells, each carrying one φ-bit of ledger information (one independent recognition event). The number of independent cells is $N = A/(4\ell_P^2)$. Each cell contributes one bit of von Neumann entropy to an external observer who cannot access the interior double-entry partner. The standard thermodynamic derivation, $dS_{\text{BH}} = dM/T_H$, with $T_H = \hbar c^3/(8\pi G M k_B)$, integrates to

$S_{\mathrm{BH}} = A/(4\ell_P^2)$ (in natural units with $k_B = 1$). The factor 1/4 encodes the specific double-entry geometry of the ledger; the cell-counting argument gives the correct parametric dependence $S \propto A/\ell_P^2$ and the correct order of magnitude. □

Remark 6.2 (Proof-sketch caveat). The cell-counting argument is a proof of proportionality, not of the precise coefficient 1/4. In standard semiclassical gravity, the 1/4 coefficient is derived by integrating $dS = dM/T_H$ from $M = 0$ to M; in loop quantum gravity, it fixes the Barbero–Immirzi parameter. In RS, the coefficient 1/4 is expected to follow from the double-entry normalization of the φ-bit counting on the horizon lattice, but the detailed combinatorial argument is an identified open problem. The formula $S_{\mathrm{BH}} = A/(4\ell_P^2)$ is reproduced correctly by the thermodynamic route (Theorem 7.1 + first law of black hole mechanics) and is taken as exact.

Corollary 6.3 (Bekenstein Bound)**.** *The entropy of any region of radius R containing energy E is bounded by $S \leq 2\pi RE/(\hbar c)$.*

Remark 6.4. The area law (not volume law) for black hole entropy is natural in RS: the ledger entries accessible to an external observer are those on the causal boundary (the horizon), not those in the interior. The entanglement entropy between interior and exterior events is proportional to the boundary area, matching the general area law for entanglement entropy in the ledger.

7 Hawking Temperature from J-Cost

Proposition 7.1 (Hawking Temperature — Structural Derivation)**.** *The Hawking temperature of a black hole of mass M is structurally identified as:*

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}. \tag{4}$$

Structural derivation. The surface gravity of a Schwarzschild black hole is $\kappa = c^4/(4GM)$ (standard GR result [3]). The minimum J-cost excitation at the horizon has energy $E_{\min} = \hbar\kappa/(2\pi c)$. Identifying the thermal energy $k_B T_H = E_{\min}$ gives $T_H = \hbar\kappa/(2\pi c k_B) = \hbar c^3/(8\pi G M k_B)$. This is a structural identification: the surface gravity is taken from standard GR and the identification $k_B T_H = E_{\min}$ is the RS restatement of the Unruh effect at the horizon. A derivation of κ from the RS Hamiltonian (rather than from standard GR) is an identified open problem; see also the companion paper on Hawking Radiation Temperature. □

Remark 7.2. As M decreases through evaporation, T_H increases. The final stages of evaporation reach $T_H \sim T_{\mathrm{Planck}}$, where the discrete ledger structure becomes dominant and the semiclassical approximation breaks down.

8 Where the Semiclassical Approximation Fails

Proposition 8.1 (Origin of Apparent Information Loss — Structural Argument)**.** *Hawking's information loss is an artifact of the semiclassical approximation, which replaces the discrete ledger with a smooth manifold.*

Proof. Hawking's calculation uses quantum field theory on a fixed classical background $g_{\mu\nu}$. This approximation:

(i) Replaces the discrete ledger (integer-valued entries, double-entry pairs) with a smooth manifold (continuous fields).

(ii) Discards the double-entry correlations between interior and exterior events.

(iii) Produces a thermal state $\rho_{\text{rad}} \propto e^{-\hbar\omega/(k_B T_H)}$, which is the maximum-entropy state consistent with the smooth background—not the full quantum state.

When the discrete ledger structure is retained, the correlations between interior events $\mathcal{L}_{\text{in}}$ and exterior events $\mathcal{L}_{\text{out}}$ encode exactly the information that the smooth approximation discards. The full state is pure; the thermal appearance arises from tracing over $\mathcal{L}_{\text{in}}$. (The claim that the full state is pure and information is recovered is the RS structural assertion; the full dynamical demonstration that the double-entry correlations encode the initial state—equivalent to computing sub-leading Hawking radiation corrections—is an identified open problem.) □

9 The Page Curve

Page [4] showed that if evaporation is unitary, the entanglement entropy S_{ent} between the radiation and the remaining black hole must follow a specific curve: rising until the Page time t_P (when approximately half the initial entropy has been radiated), then falling to zero as the black hole evaporates completely.

Proposition 9.1 (RS Page Curve — Structural Argument)**.** *The RS ledger reproduces the qualitative features of the Page curve.*

Proof. Let $n_{\text{in}}(t)$ and $n_{\text{out}}(t)$ be the number of ledger entries inside and outside the horizon at tick t. At $t = 0$, $n_{\text{out}} = 0$ and $S_{\text{ent}} = 0$.

Phase I ($t < t_P$): Each Hawking emission creates an entangled pair—one entry escapes to $\mathcal{L}_{\text{out}}$ while its double-entry partner remains in $\mathcal{L}_{\text{in}}$. The entanglement entropy grows as $S_{\text{ent}} \approx n_{\text{out}} \cdot k_B \ln 2$, increasing with each emission.

Page time ($t = t_P$): When $n_{\text{out}} \approx n_{\text{in}}$, the entanglement entropy is maximal: $S_{\text{ent}}^{\max} \approx S_{\text{BH}}/2$.

Phase II ($t > t_P$): The shrinking interior has fewer entries to entangle with. Each subsequent emission draws from the remaining $\mathcal{L}_{\text{in}}$, and the radiation begins to purify through the double-entry correlations. S_{ent} decreases monotonically.

Final state: When $n_{\text{in}} = 0$ (complete evaporation), all double-entry pairs have been transferred to $\mathcal{L}_{\text{out}}$. $S_{\text{ent}} = 0$: the radiation is a pure state. □

Remark 9.2. This argument reproduces the qualitative shape of the Page curve—growing then decreasing entanglement entropy—using the double-entry pairing mechanism. A quantitative calculation of the entanglement entropy (including the precise location of the Page time and sub-leading corrections) from the RS ledger dynamics is an identified open problem. This is consistent with the quantum extremal surface calculations [8] that recover the Page curve via the island formula; RS provides a candidate microscopic mechanism (double-entry pairing) for the island.

10 Singularity Resolution

Proposition 10.1 (Singularity Resolution)**.** *The classical Schwarzschild singularity ($r = 0$, infinite curvature) does not exist in RS. The discrete ledger provides a natural UV cutoff at the Planck scale.*

Proof. The curvature R in RS is the second variation of the J-cost density. Three properties bound it:

(i) $J(x) \geq 0$ with minimum at $x = 1$ (Lemma 2.4), so the cost density is non-negative everywhere.

(ii) The ledger has a minimum voxel size $\ell_0 = c\tau_0$, where τ_0 is the tick duration. Below this scale, the continuum description breaks down.

(iii) The maximum curvature is bounded by the cost curvature at the minimum: $|R| \leq J''(1)/\ell_0^2 = 1/\ell_0^2 < \infty$.

The classical singularity arises from extrapolating GR's smooth manifold below the Planck scale. The discrete ledger structure replaces the singular point with a Planck-density core of finite curvature and finite J-cost. □

Remark 10.2. This is consistent with loop quantum gravity [10], where quantum geometry effects resolve the singularity via a "bounce." In RS, the resolution is more fundamental: the continuum manifold itself is an approximation to the discrete ledger, and the singularity is an artifact of the approximation.

11 Comparison with Other Resolutions

Resolution	**Info fate**	**Unitary?**	**Firewall?**	**Singularity?**	**Free params**
Hawking (1976)	Lost	No	No	Yes	0
Remnant	Stored	Yes	No	Maybe	0
Complementarity	Duplicated	Yes	No	Yes	0
Firewall (AMPS)	Lost at wall	Yes	Yes	Yes	0
ER=EPR	Geometric	Yes	No	Unclear	0
Island formula	Encoded	Yes	No	Unclear	0
RS (ledger)	**Conserved**	**Yes**	**No**	**No**	**0**

12 Predictions and Falsifiers

1. **Unitarity**: No genuine information loss in any gravitational process. Any future experiment demonstrating unitarity violation would falsify RS.

2. **Page curve**: The entanglement entropy of Hawking radiation follows the Page curve, peaking at half the initial Bekenstein–Hawking entropy.

3. **No firewall**: An infalling observer experiences nothing unusual at the horizon (smooth crossing). The horizon is a causal boundary, not a physical surface.

4. **No singularity**: The classical $r = 0$ singularity is replaced by a Planck-density core with finite curvature.

5. **Area-entropy**: $S_{\mathrm{BH}} = A/(4\ell_P^2)$ exactly, with corrections $\sim \ln A$ from discrete ledger effects.

6. **Falsifier**: Detection of Hawking radiation with demonstrably zero correlations (truly thermal to all orders) would falsify the ledger-correlation mechanism.

13 Conclusion

The black hole information paradox is resolved by three structural properties of the recognition ledger: (i) double-entry bookkeeping from reciprocal symmetry $J(x) = J(x^{-1})$ conserves all ledger entries; (ii) the append-only no-sink theorem prohibits information deletion; (iii) the horizon is a causal boundary, not an information barrier. The Bekenstein–Hawking entropy counts φ-bits on the horizon surface, the Hawking temperature is the minimum J-cost excitation, and the Page curve follows from the double-entry pairing structure. No firewall, no remnant, no singularity, no free parameters.

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